THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET B

Chapters 9 & 10: **The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The diagonal of a square of side 10 cm is: 1

(a) 20 cm **(b)** $10\sqrt{2}$ cm **(c)** $10\sqrt{3}$ cm **(d)** 10 cm

2. The side of a square whose area is double that of a square of side a is: 1

(a) $a + 2$ **(b)** $a\sqrt{2}$ **(c)** $a\sqrt{3}$ **(d)** $2a$

3. If $a^2 + b^2 < c^2$ in a triangle, the triangle is: 1

(a) equilateral **(b)** acute-angled **(c)** obtuse-angled **(d)** right-angled

4. A ladder 10 m long reaches 8 m up a wall. The foot of the ladder is how far from the wall? 1

(a) 4 m **(b)** 2 m **(c)** 8 m **(d)** 6 m

5. The distance between points (0, 0) and (5, 12) is: 1

(a) 12 **(b)** 13 **(c)** 7 **(d)** 17

6. In a pie chart, a sector representing 30% of the data has a central angle of: 1

(a) 118° **(b)** 30° **(c)** 108° **(d)** 60°

7. A pie chart is made from a circle, whose total angle is: 1

(a) 90° **(b)** 100° **(c)** 360° **(d)** 180°

8. If 10 pipes fill a tank in 6 hours, 15 pipes will fill it in: 1

(a) 9 hours **(b)** 3 hours **(c)** 4 hours **(d)** 5 hours

9. The sector angle for 90° of a pie chart represents what percent? 1

(a) 10% **(b)** 75% **(c)** 25% **(d)** 50%

10. The slope of a speed–time reasoning: doubling men halves days. This is: 1

(a) inverse proportion **(b)** direct proportion

(c) equal ratio **(d)** no relation

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Write two triplets in which the smallest member is even and one with smallest member odd. 2

12. What does Baudhayana's rectangle-diagonal rule say? 2

13. The ratio of lengths on a blueprint is 1 : 40. A room 6 m long appears how long on it? 2

14. Find the missing term: 3 : 4 : 5 = 9 : x : y. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Using $\left(n,\ \frac{n^2-1}{2},\ \frac{n^2+1}{2}\right)$ find the triplet for $n = 5$ and verify. 3

16. 8 workers can complete a task in 45 days. How many days will 18 workers take, and how many extra days if 8 had worked? 3

17. Using $\left(n,\ \frac{n^2-1}{2},\ \frac{n^2+1}{2}\right)$ find the triplet for $n = 7$ and verify. 3

OR

Using $\left(n,\ \frac{n^2-1}{2},\ \frac{n^2+1}{2}\right)$ find the triplet for $n = 9$ and verify. 3

18. In a survey of 240 people, the pie chart angles for tea, coffee and juice are 150°, 120° and 90°. Find the number for each. 3

OR

A and B invest ₹ 12000 and ₹ 18000. They share a profit of ₹ 7500. Find each share. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Find the height of an isosceles trapezium with parallel sides 14 cm and 24 cm and slant sides 13 cm each. 5
(ii) Find its area.

OR

(i) State the Shulba rule for the diagonal of a rectangle. 5
(ii) Use it to find the diagonal of a 12 by 35 rectangle.

20. (i) A recipe for 4 uses 300 g flour. How much for 10? 5
(ii) Which type of proportion?

OR

Case study: The population of a town is in the ratio 3 : 2 : 5 among children, adults and seniors; total 20000. 5
(i) Find each.
(ii) Draw the pie chart angles.
(iii) What percent are seniors?

END OF QUESTION PAPER — Total: 40 Marks

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